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Pulse shape analysis of signals from SiPM-based CsI(Tl) detectors for low-energy protons: Saturation correction and particle identification

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ABSTRACT

CsI(Tl) scintillation detectors based on a silicon photomultiplier (SiPM) were tested for low-energy proton detection. Pulse shape analysis of the SiPM signals with a transimpedance amplifier was conducted to investigate the saturation effect due to the recovery time of a pixel and to evaluate particle identification capability. To recover energy linearity, a saturation correction method based on a non-paralyzable dead time model was proposed. The setup using the SiPM with a 25 μm pixel pitch well separated protons and deuterons at energies from 3 to 12 MeV. Additionally, a readout circuit without an amplifier was tested to investigate another type of saturation due to the voltage drop effect.

1. Introduction

Our research aims to develop simple detection and analysis methods for light recoil charged particles produced in nuclear reactions with low-energy radioactive ion beams in inverse kinematics. Typically, recoil protons from resonant elastic scattering in inverse kinematics [1] have energies below 20 MeV.

CsI(Tl) scintillation detectors are sometimes used for detecting charged reaction products because they are low-cost alternatives to silicon semiconductor detectors. A photodiode (PD) is often used to detect CsI(Tl) scintillation light because the PD spectral sensitivity is a better match with the CsI(Tl) light spectrum than that of a photomultiplier tube (PMT). However, a high-gain charge sensitive amplifier is needed. This is a major drawback of PDs with respect to PMTs, which do not require amplifiers.

In the past 15 years, diverse applications have employed silicon photomultipliers (SiPMs) [2,3]. A SiPM, which consists of a large number of Geiger-mode avalanche photodiode (GAPD) micropixels, has a similar photo spectral sensitivity as a PD. It can be operated with a low-gain amplifier such as a transimpedance amplifier [4], and in some cases, an amplifier is not necessary. Additionally, a SiPM can be operated at a relatively low voltage. Therefore, SiPMs may be alternatives to PDs for CsI(Tl).

However, fundamental studies must be conducted before developing realistic detector designs. This study focuses on understanding the characteristics of SiPM signals when a SiPM is combined with a CsI(Tl) scintillator for low-energy protons.

One concern is saturation in the SiPM output. Saturation occurs when numerous incident photons (e.g., a comparable number to that of GAPD pixels) produce avalanche discharge pulses in a period shorter than the pixel recovery time. Unlike X-rays, gamma-rays, or high-energy minimum ionizing particles, a proton at several MeV or higher produces numerous photons. Thus, the SiPM saturation is non-negligible. Herein, we report the saturation effect in SiPM signal waveforms for protons and use pulse shape analysis to correct it.

If a SiPM operates without an amplifier and a shunt resistor reads out its current, the current may contain an additional saturation effect due to the voltage drop at the shunt resistor. Herein we show the voltage drop effect and its correction.

The time distribution $L(t)$ of CsI(Tl) scintillation photons has several decay components. It can be expressed as $L(t) = \sum_i (h_i/\tau_i)\exp(-t/\tau_i)$, where h_i is the total number of photons attributed to the i-th component and τ_i is its decay time constant. $L(t)$ can be approximated by two major (fast and slow) components, and the ratio $h_{\mathrm{slow}}/(h_{\mathrm{fast}}+h_{\mathrm{slow}})$ depends on the particle species. Although many previous works on particle identification (PID) have utilized this characteristic (e.g., Refs. [5, 6]), few have investigated the PID resolution at an extremely low energy close to 0 MeV. The PID resolution in this region tends to be worse due to the low light output. To address this shortcoming, we evaluate the PID resolution with SiPM at 3 MeV in this study.

Bondí et al. studied a large size CsI(Tl) crystal read out by SiPMs for low-energy protons from 2.5 to 24 MeV [7]. They investigated the linearity, energy resolution, and position dependence for the large crystal. In their experimental conditions, SiPM saturation was negligible due to the small light collection efficiencies. Both SiPM saturation and PID performance were beyond the scope of their study. Here, the present work uses protons and deuterons from 3 to 12 MeV to study

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Table 1
Specifications of Hamamatsu S13360 MPPCs [10].

Type no.	Pixel pitch	Number of pixels N_p	Over voltage ΔV	Gain G	Photon detection efficiency[a] ε	Crosstalk excess charge factor f_c
6025CS	25 μm	240 × 240	5 V	7.0×10^5	0.25	1.01
6050CS	50 μm	120 × 120	3 V	1.7×10^6	0.40	1.03
6075CS	75 μm	80 × 80	3 V	4.0×10^6	0.50	1.07

[a]At a wavelength of $\lambda = 450$ nm.

the characteristics of SiPM-based CsI(Tl) detectors and to test their PID performance.

2. Experiment

2.1. CsI(Tl) detector

Fig. 1 schematically depicts the detector structure. A CsI(Tl) crystal ($5.5 \times 5.5 \times 5.5$ mm^3, Saint-Gobain Crystals [8]) was used to absorb the total kinetic energy of the incident particle. The crystal was placed in a Teflon container, which served as a reflector for scintillation light. For low-energy charged particles, a thin reflective entrance window is important to minimize energy loss [9]. Hence, the entrance surface of the crystal was coated with a vapor-deposited aluminum layer that was approximately 0.1 μm thick. The SiPM was attached to the light exit surface of the crystal. A light guide was not used because the SiPM light-sensitive area (6 × 6 mm^2) was almost the same size as the light exit surface (5.5 × 5.5 mm^2). A small amount of BC-630 optical grease was placed on the interface between the crystal and the SiPM. A collimator of $\phi = 4$ mm was placed in front of the detector to eliminate the edge effect, which deteriorates the energy resolution.

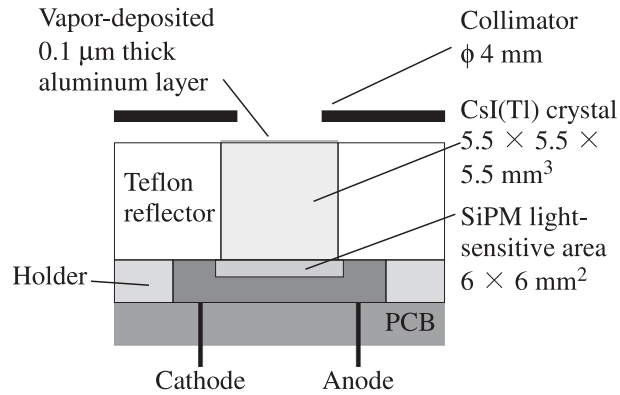

Fig. 1. Schematic structure of the SiPM-based CsI(Tl) detector.

2.2. SiPM

The SiPMs were Multi Pixel Photon Counters (MPPCs) in the Hamamatsu S13360 series [10] (Table 1). To study the saturation of the SiPM output, three types of S13360 MPPCs with different pixel pitches (25, 50, and 75 μm) were tested. All the SiPMs had a light-sensitive area of 6 × 6 mm^2. In each SiPM, GAPD pixels were two-dimensionally arranged with the pixel pitch in the square light-sensitive area. Hence, the numbers of pixels were 240 × 240, 120 × 120, and 80 × 80 for the 25, 50, and 75 μm pixel pitch SiPM, respectively.

The bias power supply voltage V_s was determined so that the gain G became the value shown in Table 1. Since G depended on the over voltage ΔV, ΔV was kept at the value shown in Table 1. Then the supply voltage was determined as $V_s = V_{br} + \Delta V$, where V_{br} is the breakdown voltage. V_s was adjusted for each measurement by considering the temperature dependence of V_{br}. Using the detector temperature T monitored by a thermistor, V_{br} was calculated as $V_{br} = V_{br}(T_0) + \alpha(T - T_0)$, where $V_{br}(T_0)$ is the factory determined breakdown voltage at a temperature of $T_0 = 25$ °C, and $\alpha = 54$ mV/°C is the temperature coefficient given in the product data sheet [10]. A typical $V_{br}(T_0)$ value was around 53 V. To maintain a constant V_s value during a measurement, the experimental room temperature was kept around 21 °C using an air conditioner.

2.3. Readout electronics

Fig. 2(a) shows the readout electronics circuit for the SiPM using a transimpedance amplifier. Unless noted otherwise, experimental data were collected using the transimpedance amplifier. When a scintillation light pulse was received by the SiPM, the current pulse $I(t)$ was drawn from the SiPM to the transimpedance amplifier. After passing through a feedback resistance $R_f = 50$ Ω, the current pulse was converted into the voltage output pulse $V(t)$ of the amplifier. This voltage pulse was transmitted through a coaxial cable with a characteristic impedance of 50 Ω, and it was observed by a digital oscilloscope (Iwatsu DS-5654) with a terminal impedance $R_t = 50$ Ω. The voltage waveform $V(t)$ was proportional to the detector current waveform $I(t)$ and was expressed as

$$V(t) = -\frac{R_f R_o}{R_o + R_t} I(t), \tag{1}$$

where $R_o = 50$ Ω is the output impedance of the amplifier. Therefore, the output polarity was negative. It should be noted that the electric potential at the amplifier input (A in Fig. 2(a)) was kept at 0 V due to the negative feedback of the amplifier circuit as this was imperative to maintain a constant SiPM bias voltage. Digitally recorded voltage waveform data were used for the pulse shape analysis.

A simple circuit without an amplifier (Fig. 2(b)) was also tested. In this circuit, the detector current $I(t)$ was shared by a shunt resistance $R_s = 50$ Ω and the terminal impedance. Therefore, the observed voltage signal was $V(t) = I(t)R_s R_t/(R_s + R_t)$. This provided essentially the same result as Eq. (1), except the output polarity of this circuit was positive.

A drawback of not using an amplifier was that the electric potential at the current exit of the SiPM (A in Fig. 2(b)) was not constant. Instead, the electric potential at the current exit was equal to $V(t)$. Therefore, the operation voltage V_{op} differed from the power supply voltage V_s, and was given as $V_{op} = V_s - V(t)$. Consequently, V_{op} varied during a scintillation pulse. If $V(t)$ is much smaller than the over voltage, the effect of the time-dependent bias voltage ("voltage drop effect") may be negligible or corrected. An advantage of a circuit without an amplifier is its simplicity as the number of attached electronic components can be minimized and an extra power supply is not required.

2.4. Beam test

The proton and deuteron beams from the 8UDH pelletron accelerator at the Kyushu University Tandem Accelerator Facility, which is part of the Center for Accelerator and Beam Applied Science of Kyushu University, were used to test three detectors with the same design as shown in Fig. 1 (pixel pitches of 25, 50, and 75 μm). Since direct beams from the accelerator were too intense for the detectors, elastically scattered particles from a 1 μm thick gold target were used. All the detectors were installed and tested simultaneously at a scattering angle of 45° in a scattering vacuum chamber. To investigate the energy dependence of the detector responses, the same energy set of 3.00, 6.00, 9.00 and 12.00 MeV was used for both scattered protons and deuterons. The electrostatic acceleration voltage was adjusted to obtain the above-specified energies for scattered particles by considering the kinematics and energy loss in the target. The beam intensities were adjusted to keep the detector count rates below 100 counts/s.

For each beam species and energy condition, approximately 4000 events of waveform data were accumulated. The waveform data were

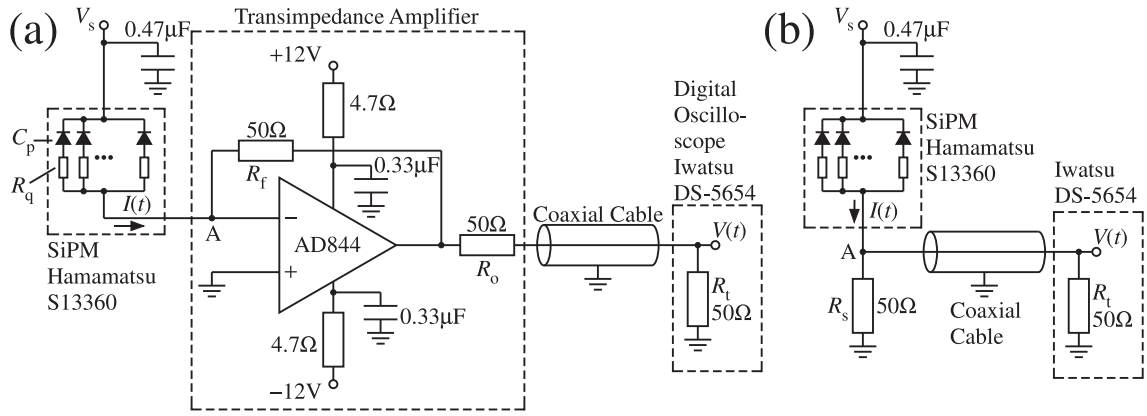

Fig. 2. Readout circuits (a) with and (b) without a transimpedance amplifier.

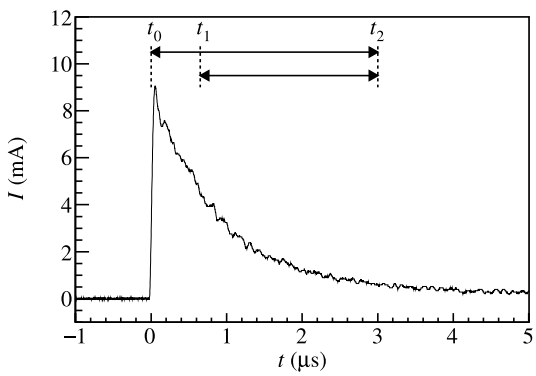

Fig. 3. Typical single-event waveform of the 25 μm pixel pitch SiPM for a 12 MeV proton. Times t_0, t_1, and t_2 define the integration gates represented by the arrows (see text).

taken with a sampling rate of 500 MHz. The trigger threshold of the digital oscilloscope was set to 10~20% of the signal pulse height for each run.

3. Pulse shape analysis and discussion

3.1. Waveform data

Eq. (1) expresses the conversion of digitized $V(t)$ waveforms to $I(t)$. Fig. 3 shows a typical single-event $I(t)$ waveform for the 25 μm pixel pitch SiPM and a 12 MeV proton. $I(t)$, which is the current drawn from the SiPM, can be regarded as the light output signal of the detector. In the present application, individual single-photon pulses could not be observed separately due to the large photon intensities. The observed waveform exhibits a typical two-component exponential decay shape for a CsI(Tl) scintillation. Fluctuations at the decaying part of the waveform are attributed to statistical fluctuations in the number of detected photons per unit time. The fluctuation at the peak ($0.05 < t < 0.2$ μs) is partly due to the response function of the amplifier, as mentioned below.

The solid curve in Fig. 4(a) represents the average waveform of the 25 μm SiPM for approximately 4000 events of 12 MeV protons. The statistical fluctuation is greatly reduced, and the intrinsic time distribution of CsI(Tl) can be observed. The small kink at the peak ($0.05 < t < 0.2$ μs) is attributed to overshooting of the op-amp used in the transimpedance amplifier.

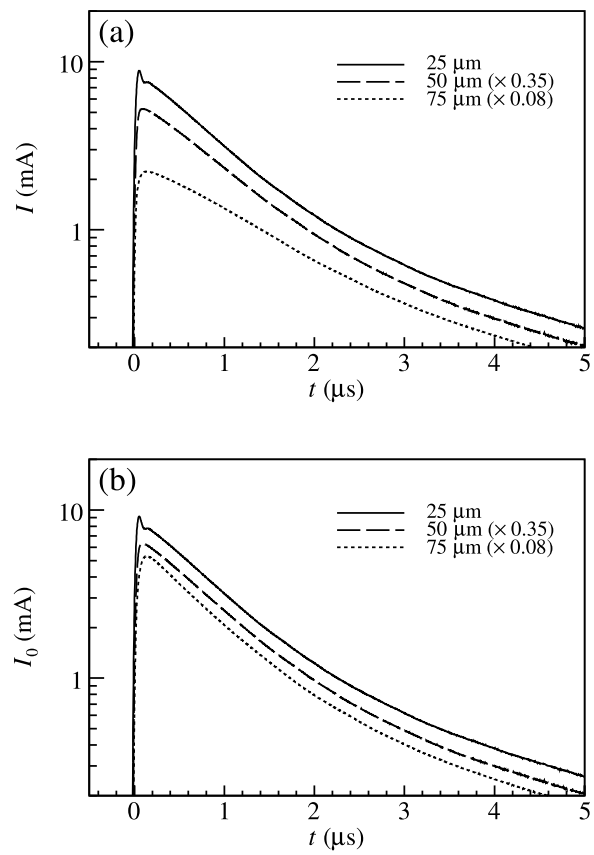

Fig. 4. Average waveforms of the 25, 50, and 75 μm pixel pitch SiPMs (solid, dashed, and dotted lines, respectively) for 12 MeV protons (a) without and (b) with saturation correction (see text). Each average waveform is determined from approximately 4000 events. For comparison, the 50 and 75 μm waveforms are vertically scaled by factors of 0.35 and 0.08, respectively.

3.2. Saturation and its correction

This subsection deals with the saturation effect in the average waveform and proposes a simple correction method. Fig. 4(a) shows the average waveforms with the 25, 50, and 75 μm pixel pitch SiPM for 12 MeV protons on a vertical logarithmic scale. In the early time region (e.g., $0.2 < t < 2$ μs) with large photon intensities, the waveforms have different decay slopes. Without a saturation effect, these three waveforms would have identical decay slopes. Assuming that the SiPMs

Table 2
Effective recovery time τ_r and expected recovery time $\tau_0 = C_p R_q$ evaluated for the 25, 50, and 75 μm pixel pitch SiPMs. C_p is the capacitance of a pixel. R_q is the resistance of a quenching resistor (see text).

Pixel pitch (μm)	τ_r (ns)	τ_0 (ns)	C_p (pF)	R_q (kΩ)
25	20	13	0.022	600
50	37	27	0.091	300
75	73	55	0.21	260

have the same incident photon rates, the difference in the decay slopes is attributed to the pixel pitch dependence of the saturation effect.

The pixel pitch dependence of the saturation effect can be explained as follows. When photons randomly arrive at the SiPM surface with an average number of n_{ph} photons per unit time, the average event rate n of the avalanche discharge at a pixel without the saturation effect is $n = n_{ph}\varepsilon f_c f_a/N_p$, where N_p is the number of pixels, ε is the photon detection efficiency, and f_c and f_a are the excess charge factors due to crosstalk and afterpulsing, respectively. Nominal values of N_p, ε, and f_c are shown in Table 1. Although f_a values are not given in the data sheet [10], they are typically several % larger than unity [11]. If n is comparably large to the inverse of the recovery time, the saturation effect becomes significant. Due to the pixel pitch dependences of N_p, ε, and f_c (Table 1), the larger the pixel pitch, the larger n. Consequently, the saturation effect becomes larger with pixel pitch.

The proposed saturation correction method below estimates n from the average number of detected photons x per pixel per unit time. Here, we consider a non-paralyzable (non-extendable) dead time model [12] for a pixel. After an initial single photon is detected at a given pixel, it is inactive for a recovery time τ_r. The total dead time per unit time at the pixel is $\tau_r x$. During the total dead time, $n\tau_r x$ events are lost out of n events. Therefore, $x = n(1-\tau_r x)$, or equivalently $n = x/(1-\tau_r x)$. The actual average current of SiPM can be factorized as $I = eGN_p x$, where e is the elementary charge. Correspondingly, the saturation corrected average current (I_0) is $I_0 = eGN_p n$. Therefore, the correction formula is given as

$$I_0(t) = \frac{I(t)}{1-\tau_r I(t)/(eGN_p)}. \quad (2)$$

In general, the parameter τ_r must be determined using calibration data with a known time distribution of photons.

It should be noted that the actual GAPD pixel during the recovery time is partially active with the time-dependent efficiency and gain [13]. A more realistic correction formula, which considers this fact, may be employed (e.g., Eq. 20 in Ref. [13]). However, such a formula can be reduced to Eq. (2) when the saturation effect is not too large ($\tau_r I/(eGN_p) \ll 1$) [13].

It should also be noted that the actual photon intensity is not constant. Instead, it decreases with the characteristic time constants of CsI(Tl). However, Eq. (2), which is based on constant intensities, can approximate the correction in the present conditions where the decay time constants are at least 10 times larger than τ_r. The effectiveness of the correction is discussed below.

In Eq. (2), G is fixed to the nominal value shown in Table 1. On the other hand, N_p is fixed to 84% of the nominal value in Table 1 because the crystal surface area was 84% of the light-sensitive area.

The τ_r parameter was determined for each SiPM by fitting. The saturation corrected $I_0(t)$ may follow a model light curve of $I_{model}(t) = I_1\exp(-t/\tau_1) + I_2\exp(-t/\tau_2) + I_{off}$, where the first two terms represent the fast and slow components of the CsI(Tl) scintillation, while I_{off} is a small offset, which includes the slower components and the detector dark current. Firstly, the average waveform $I(t)$ of the 25 μm SiPM for 12 MeV protons was fitted by the model light curve and the relation of Eq. (2), where I_1, τ_1, I_2, τ_2, I_{off}, and τ_r are free parameters. The fit region ranged from $t = 0.2$ to 10 μs, excluding the rising and the overshooting parts of the waveform. The fitting results are $\tau_1 = 0.75$ μs, $\tau_2 = 2.8$ μs, and $\tau_r = 20$ ns. These τ_1 and τ_2 values are consistent with previous studies [6,8]. The τ_r result will be discussed later. Secondly, the 12 MeV proton waveforms with the 50 and 75 μm SiPMs were fitted to determine their τ_r parameters. For consistency, τ_1, τ_2 and the ratio I_1/I_2 were fixed in the fitting to those determined for the 25 μm SiPM. Thus, $\tau_r = 37$ and 73 ns for the 50 and 75 μm SiPMs, respectively.

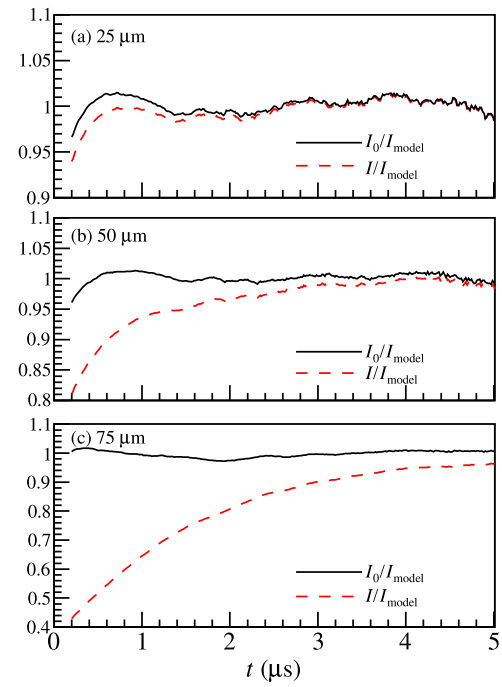

Fig. 5. I_0/I_{model} (solid line) and I/I_{model} (dashed line) ratios for the (a) 25 μm, (b) 50 μm, and (c) 75 μm pixel pitch SiPMs (see text).

Fig. 4(b) shows the saturation corrected average waveforms I_0 using the determined τ_r parameters. The differences in the decay slopes among the corrected waveforms are smaller than those among the uncorrected waveforms. Figs. 5(a)–(c) plot the I_0/I_{model} ratios for the 25, 50, and 75 μm SiPMs to verify the fit qualities. The absolute deviations of I_0/I_{model} from unity are within 3.4%, 3.9%, and 2.9% for the 25, 50, and 75 μm SiPMs, respectively, in the time region from $t = 0.2$ to 5.0 μs. For comparison, we also plotted the ratios of the uncorrected current I to I_{model}. The maximum absolute deviations of I/I_{model} from unity are 6.0%, 19%, and 57% for the 25, 50, and 75 μm SiPMs, in the same region respectively. Therefore, the saturation correction reduces the deviations from the model light curve for all SiPMs.

Each determined τ_r parameter in Table 2 should be referred to as the "effective" recovery time since it is just a parameter in the proposed fitting model and not necessarily equal to the true recovery time. For comparison, the expected recovery time τ_0 was calculated as $\tau_0 = C_p R_q$ [2,3], where C_p is the capacitance of a pixel and R_q is the resistance of a quenching resistor connected to the pixel. (See Fig. 2(a).) Note that C_p is the sum of the depletion layer capacitance and the parasitic capacitance of the quenching resistor. The C_p value of each SiPM type was estimated from the nominal values of G and ΔV in Table 1 using the relation $C_p = eG/\Delta V$ [2,3]. According to the manufacturer, the R_q values at $T = 25$ °C for the 25, 50, and 75 μm SiPMs are 600 kΩ, 300 kΩ, and 260 kΩ, respectively, with a precision of ±20% [14]. It should be noted that the quenching resistor is the same metal type as that reported in Ref. [11]. The temperature coefficient of R_q in Ref. [11] is approximately −0.1%/°C at $T = 25$ °C. Therefore, we neglected the temperature dependence of R_q, even though the present measurements were performed around $T = 21$ °C. The τ_r values are larger than the corresponding τ_0 values by factors of 1.3–1.5 (Table 2).

Table 3
Proton energy resolution of the CsI(Tl) detector with the 25 μm pixel pitch SiPM.

Energy (MeV)	3	6	9	12
RMS/Mean (%)	1.7	1.4	1.2	1.0

Table 4
Energy nonlinearity error $|\delta E_p|_{\max}$ of the 25, 50, and 75 μm pixel pitch SiPMs for protons in the energy region of 3–12 MeV without (w/o) and with (w/) the saturation correction.

Pixel pitch	$\|\delta E_p\|_{\max}$ (keV)	
(μm)	w/o	w/
25	15	3
50	120	58
75	280	55

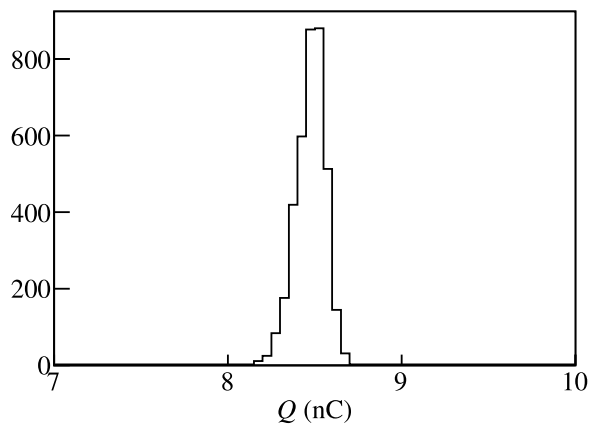

Fig. 6. Integrated charge histogram for 12 MeV protons with the 25 μm pixel pitch SiPM after saturation correction.

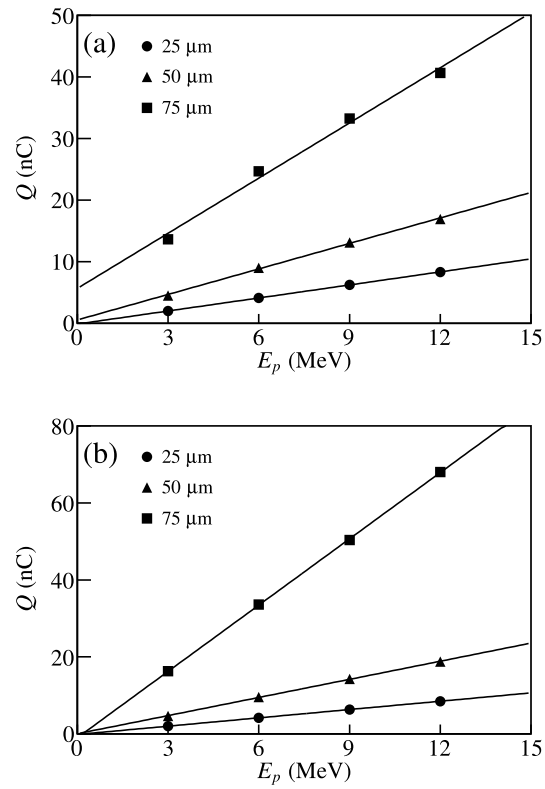

Fig. 7. Mean integrated charge Q measured by the 25, 50, and 75 μm pixel pitch SiPMs (circles, triangles, and squares, respectively) as functions of incident proton energy E_p (a) without and (b) with saturation correction. Solid lines represent the results of fits by a straight line.

3.3. Integrated charge

Here we discuss the energy resolution and energy linearity of the detectors. Integrating a single-event waveform $I(t)$ over a time gate from $t_0 = 0$ μs to $t_2 = 3$ μs extracted energy information on an event-by-event basis (Fig. 3). Although the t_2 value was chosen to provide the best energy resolution, the energy resolution did not depend significantly on t_2 for $t_2 > 1$ μs. The integration result Q gives the total charge produced during the time window, which should be proportional to the kinetic energy of the incident particle.

We also applied the above saturation correction to single-event waveforms to improve the energy linearity. It should be noted that the above saturation correction method is, strictly speaking, not applicable to a single-event waveform since its $I(t)$ does not precisely represent the average photon intensity. However, the "saturation corrected" single-event waveforms using Eq. (2) can provide useful energy information because the average waveform is approximately the same as the saturation corrected average waveform discussed in the previous subsection.

Fig. 6 shows the Q histogram for 12 MeV protons with the 25 μm pixel pitch SiPM after the saturation correction. For the peak in the Q histogram, the energy resolution was evaluated as the root mean square (RMS) to mean ratio. Table 3 shows the energy resolutions of the 25 μm SiPM, which provided the best resolutions compared to the other SiPMs, for protons at 3, 6, 9, and 12 MeV.

To verify the saturation effect in Q, Figs. 7(a) and (b) plot the mean Q values without and with the saturation correction, respectively, as functions of the proton energy E_p. As expected, the 75 μm SiPM has the largest Q saturation effect. For example, the uncorrected Q of the 75 μm SiPM at 12 MeV in Fig. 7(a) is 40% smaller than the corresponding corrected Q in Fig. 7(b). On the other hand, the Q reductions for the 50 and 25 μm SiPM at 12 MeV are 10% and 1.7%, respectively. The 25 μm SiPM is the best in terms of the smallness of saturation because its Q reductions are 0.5%, 0.9%, 1.3%, and 1.7% at 3, 6, 9, and 12 MeV, respectively. By extrapolating this reduction trend with a linear function of energy, a reduction of approximately 3% should be realized at 20 MeV, which is still a small level.

To evaluate the nonlinearity, an "energy calibration" fitting was performed for each SiPM with a linear function $E_p = aQ + b$, where a and b are free parameters. After the fitting, the largest absolute residual $|\delta E_p|_{\max}$ was selected as a measure of the nonlinearity error. Table 4 shows the results of $|\delta E_p|_{\max}$ for each SiPM with and without the saturation correction. The saturation correction reduces the nonlinearities for all SiPMs, indicating that the proposed correction method is suitable.

The a parameters for the 25, 50, and 75 μm SiPM are 1.4, 0.64, and 0.17 MeV/nC, respectively, after the saturation correction. The differences among these parameters are mainly due to the differences in the SiPM parameters (i.e., G, ε, and f_c parameters in Table 1). However, using only the SiPM parameters did not precisely reproduce the ratio of the a parameters since the light collection efficiencies of the three detectors were unknown.

3.4. Voltage drop correction for the circuit without an amplifier

Waveforms with the simple circuit shown in Fig. 2(b) without an amplifier were also inspected. Fig. 8 shows the average waveforms of the 25 μm pixel pitch SiPM for 12 MeV protons read out by circuits without and with the transimpedance amplifier ($I_{\mathrm{wo}}(t)$ and $I(t)$, respectively). The decaying parts ($t > 0.3$ μs) of $I_{\mathrm{wo}}(t)$ and $I(t)$ have different decay slopes. As discussed below, this can be understood by the reduction of I_{wo} due to the voltage drop at the shunt resistance R_s. (See Fig. 2(b).)

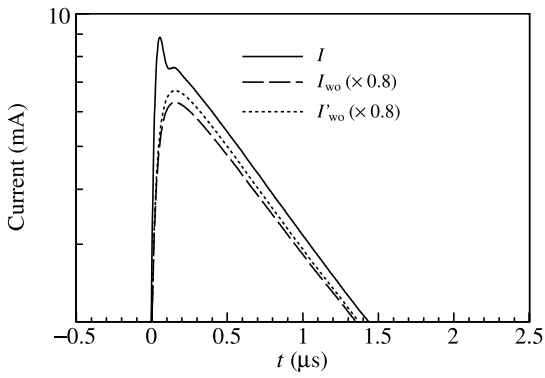

Fig. 8. Average waveforms of the 25 μm pixel pitch SiPM for 12 MeV protons read out by the circuits with (solid line) and without a transimpedance amplifier (dashed line). Dotted line shows the voltage drop corrected waveform without an amplifier (see text). Dashed and dotted waveforms are both vertically scaled by 0.8.

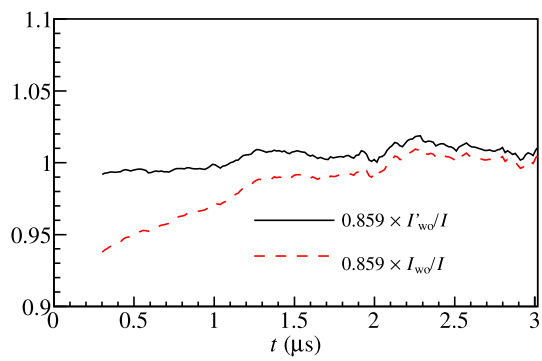

Fig. 9. I'_{wo}/I (solid line) and I_{wo}/I (dashed line) ratios for the 25 μm pixel pitch SiPM (see text). Ratios are vertically scaled by 0.859.

The reduction of I_{wo} is attributed to the over voltage (ΔV) dependences of G, ε, f_c, and f_a of the SiPM. The ΔV dependence of G is $G = C_p \Delta V/e$ [2,3] and that of ε is approximately $\varepsilon(\Delta V) = \varepsilon_{max}[1 - \exp(-(\Delta V - U_0)/U_{ch})]$ [13], which reproduces the ε graph in the data sheet [10] with a parameter set of $\varepsilon_{max} = 0.32$, $U_0 = 0.70$ V, and $U_{ch} = 2.87$ V. It is assumed that the same ε function form (i.e., with the same U_0 and U_{ch} parameters) is also applicable to CsI(Tl) scintillation light. For simplicity, we neglected the ΔV dependences of f_c and f_a since they are close to unity. Therefore, the current I_{wo} is proportional to $g(\Delta V) = \varepsilon(\Delta V)\Delta V$.

The voltage-drop correction was performed using the $g(\Delta V)$ function. The over voltage with the voltage drop effect is given as $\Delta V = \Delta V_0 - R_{st} I_{wo}$, where ΔV_0 is the nominal over voltage given in Table 1 and $R_{st} = R_s R_t/(R_s + R_t)$. The dotted line in Fig. 8 shows the voltage drop-corrected waveform $I'_{wo}(t)$ realized by multiplying the measured $I_{wo}(t)$ by $g(\Delta V_0)/g(\Delta V_0 - R_{st} I_{wo}(t))$. Note that a fitting was not performed, but the decay slopes of the corrected waveform became closer to those of the waveform with the amplifier.

To evaluate the effect of the voltage drop correction, Fig. 9 plots the I'_{wo}/I and I_{wo}/I ratios. The I_{wo}/I ratio before the correction varies by approximately 6% during the time region from $t = 0.3$ to 3.0 μs. On the other hand, the I'_{wo}/I ratio after the correction remains nearly constant and the absolute deviations from the average value of $I'_{wo}/I = 0.859$ are within 2% in the same time region. Note that this average ratio differs from unity. Although different factors can affect the average ratio, we attribute this to the difference in the light collection efficiencies between the detector for the I_{wo} data and that for the I data. The present result suggests that I_{wo} is affected by the voltage drop effect. However, this effect can be avoided by employing a transimpedance amplifier. Regardless, the simple readout circuit may provide useful energy information if it is combined with the voltage drop correction.

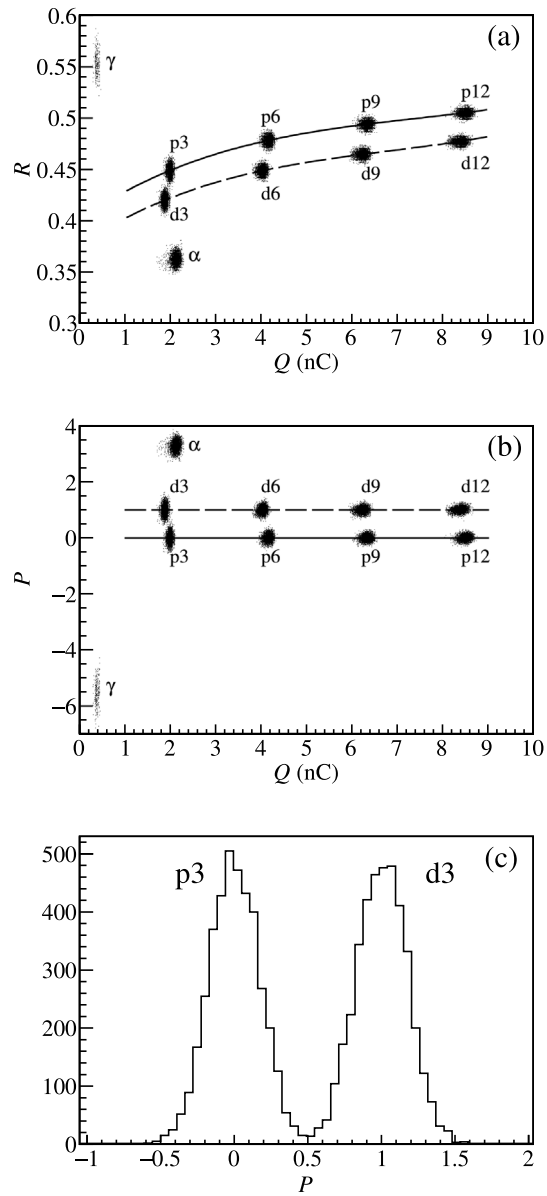

Fig. 10. (a) R vs. Q plot, (b) P vs. Q plot, and (c) P histogram derived from the particle identification analysis for the 25 μm pixel pitch SiPM (see text). Symbols p (proton) and d (deuteron) followed by energies (3, 6, 9, 12 in MeV) identify groups of events taken in separate runs of the beam test. Symbols α and γ denote events with an ^{241}Am α source and a ^{137}Cs γ source, respectively. Solid and dashed lines in the R vs. Q plot represent the polynomial functions R_p and R_d, respectively. Solid and dashed lines in the P vs. Q plot represent the $P = 0$ and $P = 1$ lines, respectively.

3.5. Particle identification

After applying the saturation correction, particle identification (PID) analysis was performed for the 25 μm pixel pitch SiPM using the ratio R of the tail part charge Q_{tail} to the total charge Q ($R = Q_{tail}/Q$). The integration time gate for Q_{tail} was set from t_1 to t_2 (Fig. 3) within the integration time gate for Q from $t_0 = 0$ μs to $t_2 = 3$ μs. The optimum value of $t_1 = 0.65$ μs was chosen as it gives the best resolution of the PID variable P, which is defined later.

Fig. 10(a) shows a two-dimensional plot of R vs. Q for events in separate runs of the beam test as well as for events with an ^{241}Am α source and a ^{137}Cs γ source. The proton and deuteron events can be

fitted with third-order polynomial functions

$$R_p(Q) = \sum_{i=0}^{3} a_i Q^i \quad \text{and} \quad R_d(Q) = \sum_{i=0}^{3} b_i Q^i, \tag{3}$$

respectively, where a_i and b_i are polynomial coefficients of the i-th order. In the energy region of 3–12 MeV, the protons and deuterons are well separated. Compared to deuteron events, α and γ events are further separated from the proton locus.

The PID variable P is defined from Q and R variables, and is given as

$$P = \frac{R - R_p(Q)}{R_d(Q) - R_p(Q)}. \tag{4}$$

$P = 0$ for protons and $P = 1$ for deuterons. P is independent of Q at least for protons and deuterons. (See the P vs. Q plot in Fig. 10(b).)

In the present energy region, the P resolution degrades as the energy decreases. The P histogram for protons and deuterons at 3 MeV was constructed to evaluate the worst-case resolution (Fig. 10(c)). Each proton and deuteron peak in the histogram was separately fitted with a Gaussian function. The proton and deuteron peaks both have widths $\sigma = 0.17$, which is small compared to the proton–deuteron separation of unity.

4. Conclusions

For high photon fluxes, a transimpedance amplifier should be used for the SiPM readout. Otherwise, the operation voltage of the SiPM depends on the instantaneous photon intensity. This leads to a nonlinear response.

The setup, which consisted of a CsI(Tl) crystal read out by SiPMs with an equal light-sensitive area, a pixel pitch of 25, 50, or 75 μm, and a transimpedance amplifier, was exposed to scattered beams of protons and deuterons with energies of 3, 6, 9, and 12 MeV as well as ^{241}Am α and ^{137}Cs γ sources. Then the current transients $I(t)$ were recorded.

The nonlinearity of the response was obtained from the integral of the transient $Q = \int I(t)\,dt$ versus the beam energy. As expected, the nonlinearity increases with increasing pixel pitch.

The correction factor for the current due to the saturation effect is modeled by $I_0(t)/I(t) = 1/(1 - \tau_r I(t)/(eGN_p))$, where $I_0(t)$ is the current corrected for the saturation effect, e is the elementary charge, G is the SiPM gain, and N_p is the number of pixels. The single free parameter is τ_r, and is interpreted as the effective recovery time that is necessary for a pixel to recover. During the recovery period, the pixel cannot generate a signal. Assuming the time dependence of the photon yield from CsI(Tl) is given as $I_1 \exp(-t/\tau_1) + I_2 \exp(-t/\tau_2) + I_{\mathrm{off}}$, the τ_r values for the different SiPMs can be determined. After the correction, the integral of the corrected transient, $Q_0 = \int I_0(t)\,dt$, versus particle energy is linear within 60 keV (0.7% of the width of the calibration range from 3 to 12 MeV) for the three SiPMs. Hence, the proposed method can correct the nonlinear response when the recovery time is much shorter than decay time constants of scintillation.

The particle identification capability of the setup using the charge fraction after 0.65 μs was also investigated. As shown for the SiPM with a 25 μm pixel pitch for protons and deuterons with energies between 3 and 12 MeV, the protons and deuterons can be separated by 5.9 σ even at the lowest energy.

Numerical data of the results presented in this paper are available at the repository site in Ref. [15].

CRediT authorship contribution statement

T. Teranishi: Conceptualization, Investigation, Writing - original draft. **Y. Ueno:** Investigation. **M. Osada:** Investigation. **S. Oka:** Investigation. **K. Iribe:** Investigation. **H. Yoshida:** Investigation. **H. Sakai:** Investigation. **T. Kubo:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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